



Analog IC Design

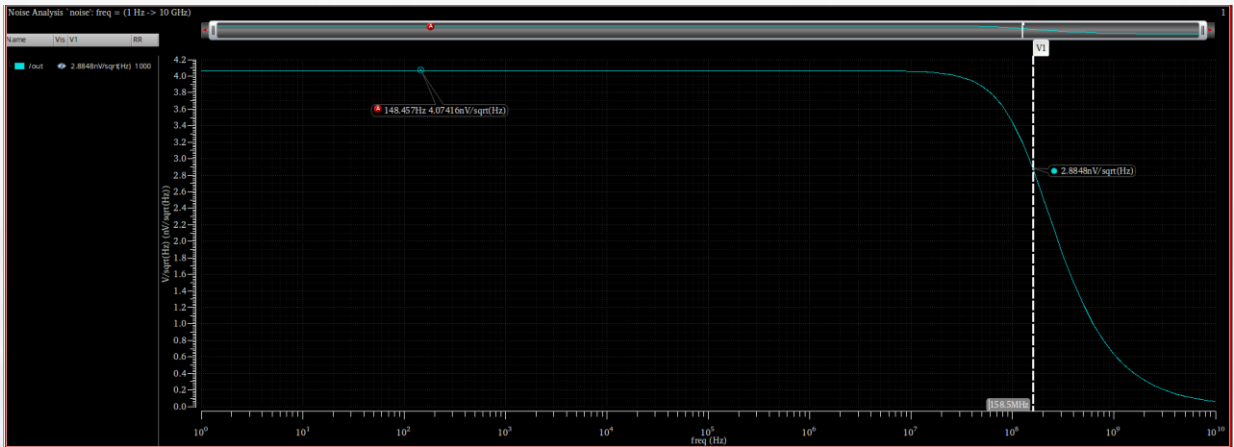
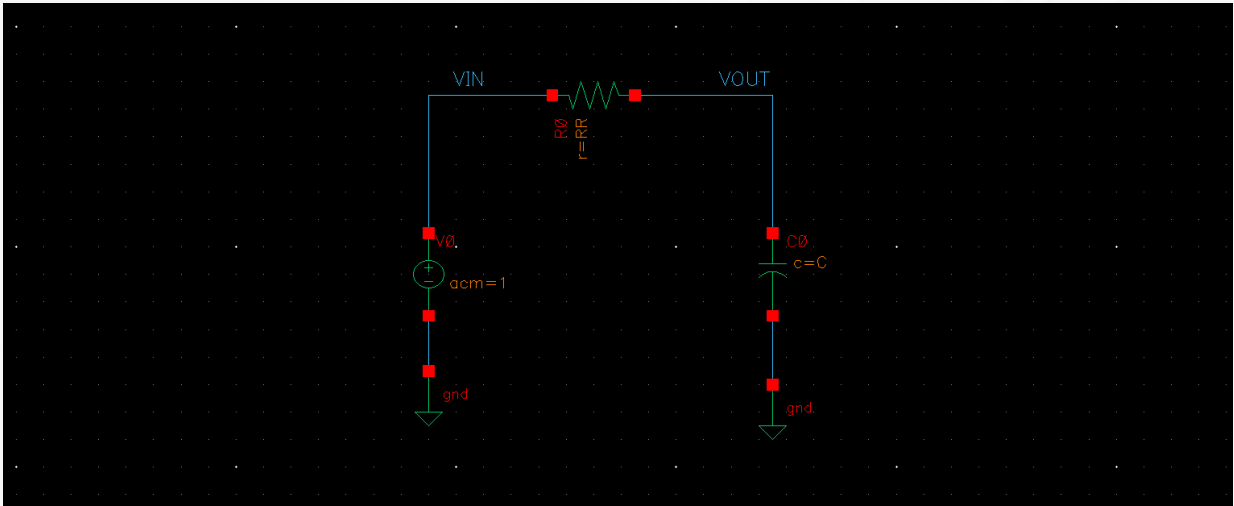
CMOS Technology

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Section 3 - Lab 10

Noise simulation

LPF AC Noise Analysis



Output noise: $\text{noise density} \approx 4.07 \text{ nV}/\sqrt{\text{Hz}}$, $BW \approx 159 \text{ MHz}$ at noise density $= \frac{4.07 \text{ nV}/\sqrt{\text{Hz}}}{\sqrt{2}} \approx 2.88 \text{ nV}/\sqrt{\text{Hz}}$

rms noise output $\approx 64.32 \mu V_{rms}$

hand calculations:

$$\text{noise density} \approx \sqrt{\frac{R}{1K\Omega}} \times 4 \text{ nV}/\sqrt{\text{Hz}} = 4 \text{ nV}/\sqrt{\text{Hz}}$$

$$BW = \frac{1}{2\pi RC} = \frac{1}{2\pi \cdot 1K\Omega \cdot 1pF} \approx 159.15 \text{ MHz}$$

$$\text{rms noise} \approx \sqrt{\frac{1pF}{C}} \times 64 \mu V_{rms} = 64 \mu V_{rms}$$

Test	Output	Nominal	Spec	Weight	Pass/Fail
lab10:RC:1	rmsNoise(1 1e+10)	64.32u			

Parameter	Simulation	Hand calculations
noise density	$4.07 \text{ nV}/\sqrt{\text{Hz}}$	$4 \text{ nV}/\sqrt{\text{Hz}}$
BW	159MHz	159.15MHz
rms noise	$64.32 \mu V_{rms}$	$64 \mu V_{rms}$

Run parametric sweep for RPAR = 1k, 10k, 100k, 1000k, Plot output noise overlaid on the same plot. Using log-scale for y-axis. Comment on the results



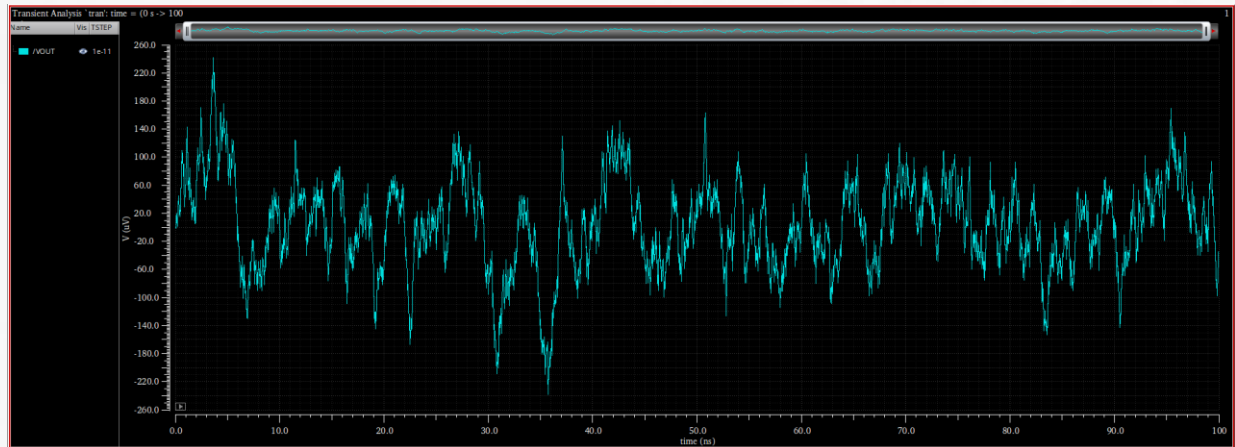
As R increases, the noise density $\propto \sqrt{R}$ increases and $BW \propto \frac{1}{R}$ decreases

Calculate the rms noise using the calculator. Comment on the results

Point	Test	Output	Nominal	Spec	Weight	Pass/Fail
Parameters: Rf=1k						
1	lab10:RC:1	rmsNoise(1 1e+10)	64.32u			
Parameters: Rf=10k						
2	lab10:RC:1	rmsNoise(1 1e+10)	64.62u			
Parameters: Rf=100k						
3	lab10:RC:1	rmsNoise(1 1e+10)	64.65u			
Parameters: Rf=1M						
4	lab10:RC:1	rmsNoise(1 1e+10)	64.66u			

As R increases, the rms noise almost remains constant as $V_{rms} \approx \sqrt{\frac{1pF}{C}} \times 64\mu V_{rms}$ almost doesn't depend on R

LPF AC Transient Noise Analysis

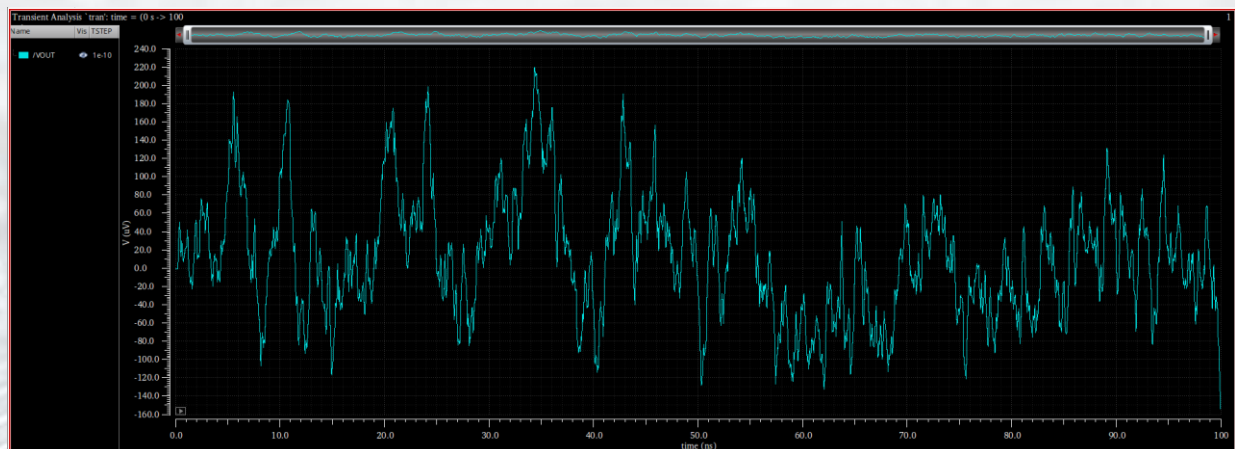


Annotate the min and max values, Use the rms function in the calculator to calculate the rms noise. Compare it to the value calculated in Part 1

Test	Output	Nominal	Spec	Weight	Pass/Fail
lab10:RC:1	rms noise	63.53u			
lab10:RC:1	min	243.3u			
lab10:RC:1	max	-237u			

Parameter	AC noise analysis	Transient noise analysis
<i>rms noise</i>	$64.32\mu V_{rms}$	$63.53\mu V_{rms}$

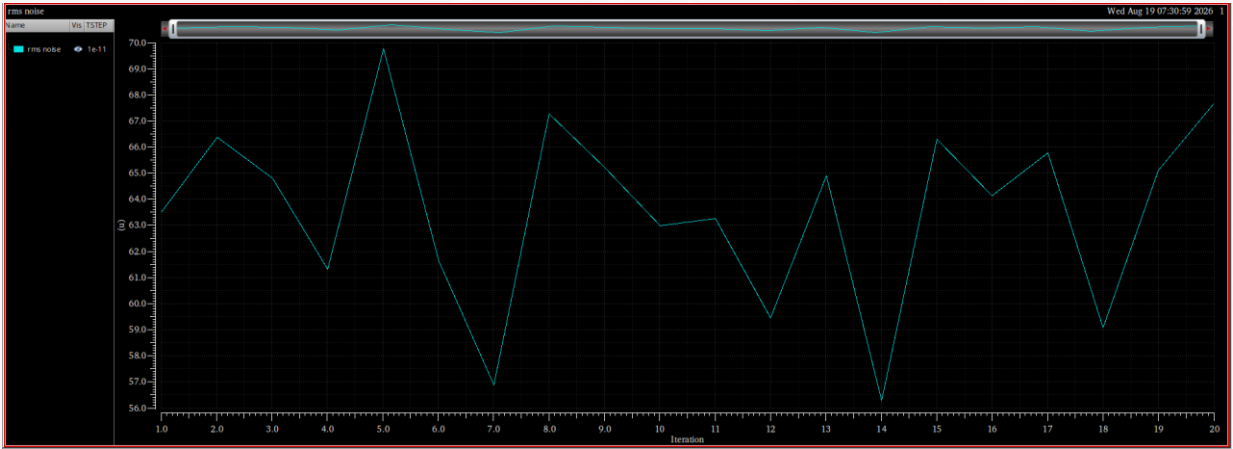
Repeat the simulation with $TSTEP = TAU/10$. Does the calculated rms noise increase or decrease? Why?



Test	Output	Nominal	Spec	Weight	Pass/Fail
lab10:RC:1	rms noise	66.57u			
lab10:RC:1	max	220.1u			
lab10:RC:1	min	-153u			

The expected form the rms noise to decrease as the frequency range decrease, but the calculated rms noise **increases**, and this is because of the randomness of the noise naturality. To get accurate results we should run the noise many times and calculate the average of the iterations.

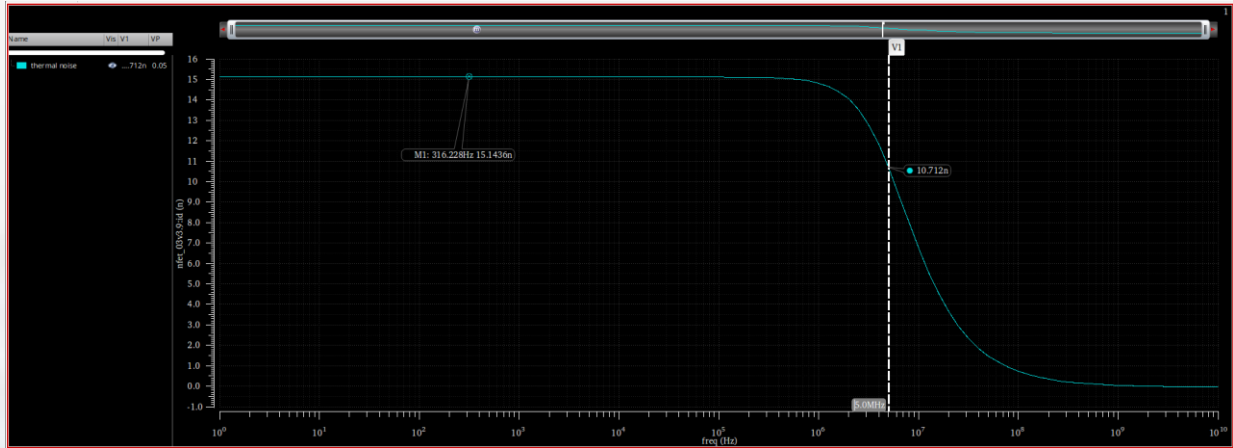
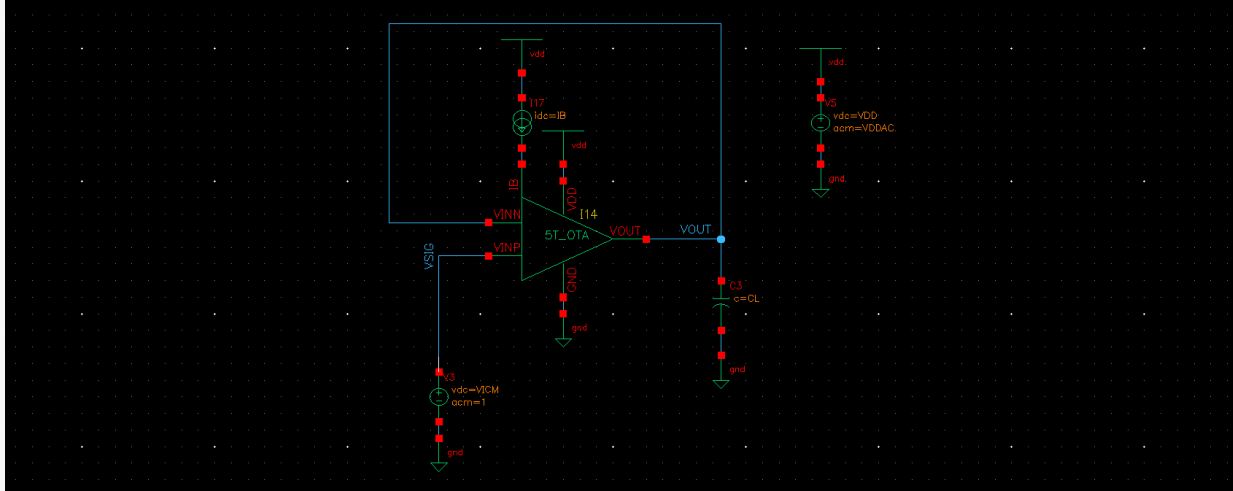
Back to TSTEP = TAU/100. Now it will run 20 runs of transient noise iterations



Test	Output	Nominal	Spec	Weight	Pass/Fail
lab10:RC:1	rms noise				
lab10:RC:1	average	63.51 u			

Parameter	AC noise analysis	Transient noise analysis	Transient noise iteration
rms noise	64.32 μV_{rms}	63.53 μV_{rms}	63.51 μV_{rms}

5T OTA AC Noise Analysis



Output thermal noise: $\text{noise density} \approx 15.15 \text{ nV}/\sqrt{\text{Hz}}$, $\text{BW} \approx 5 \text{ MHz}$ at noise density = $\frac{15.15 \text{ nV}/\sqrt{\text{Hz}}}{\sqrt{2}} \approx 10.712 \text{ nV}/\sqrt{\text{Hz}}$

Compare the simulation results with hand analysis

$$\text{Thermal noise density} \approx \sqrt{\frac{8KT\gamma}{g_{m1}} \left(1 + \frac{g_{m3}}{g_{m1}} \right)} = \sqrt{\frac{8 \times 4.14 \times 10^{-21} \text{ J} \times \frac{2}{3}}{160 \mu\text{S}} \left(1 + \frac{160 \mu\text{S}}{95 \mu\text{S}} \right)} \approx 14.83 \text{ nV}/\sqrt{\text{Hz}}$$

BW will be shaped by the overall bandwidth of the system **5MHz**

Parameter	Simulation	Hand calculations
noise density	$15.15 \text{ nV}/\sqrt{\text{Hz}}$	$14.83 \text{ nV}/\sqrt{\text{Hz}}$
BW	5MHz	5MHz

Report total output noise (thermal + flicker) vs frequency. Estimate the Flicker noise corner



■ thermal noise ■ flicker noise: $f_c \approx 1.65\text{MHz}$



■ thermal noise ■ flicker noise ■ thermal noise + flicker noise

The above graph proof that, at high frequency the thermal noise dominates and at low frequency the flicker noise dominates but around the flicker noise corner f_c the noise depends on both thermal and flicker noise.

Calculate rms output noise (calculate the rms noise due to thermal noise only using Noise Summary)

$$V_{TN_{rms}}^2 \approx (5.68 + 5.64 + 3.7 + 3.1) \times 10^{-10} + (2.82 + 1.6) \times 10^{-12} V$$

$$V_{TN_{rms}}^2 \approx 1.81 \times 10^{-9} V^2$$

$$\Rightarrow V_{TN_{rms}} \approx \sqrt{1.81 \times 10^{-9} V^2} \approx 42.55 \mu V$$

Compare the simulation results (noise density, bandwidth, and rms) with hand analysis.

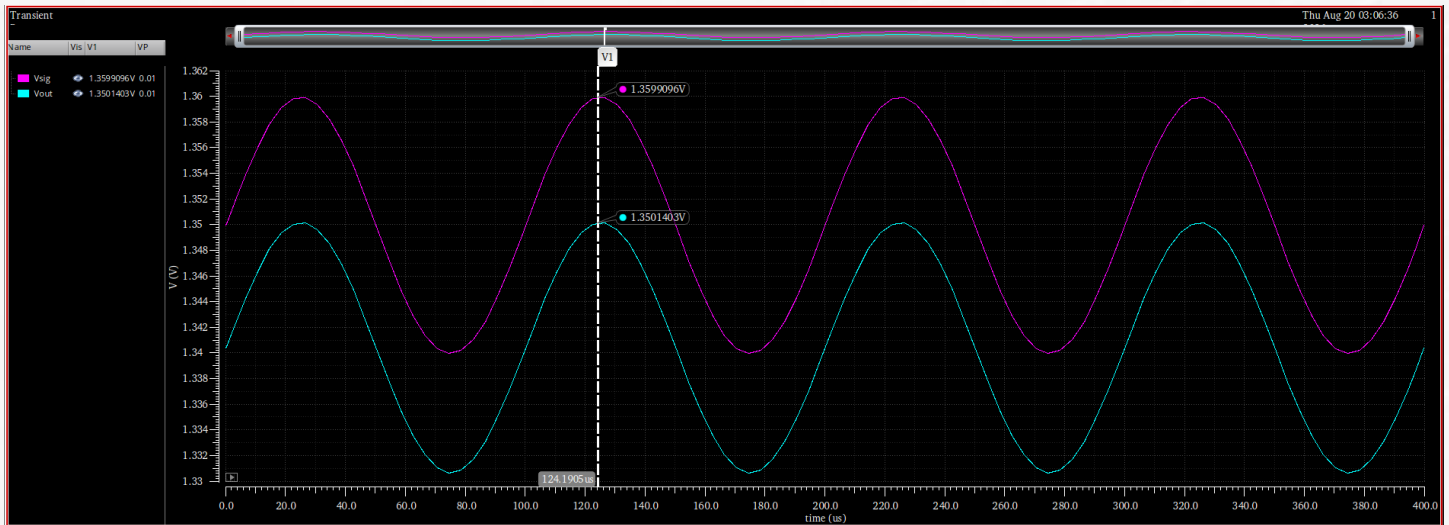
$$V_{TN_{rms}} \approx \sqrt{V_{TN}^2 \times BW} = \sqrt{(14.83 nV/\sqrt{Hz})^2 \times 5 MHz} = 41.56 \mu V_{rms}$$

Device	Param	Noise Contribution	% Of Total
I14.M2.m0	fn	2.08094e-09	30.02
I14.M1.m0	fn	1.97092e-09	28.43
I14.M2.m0	id	5.67641e-10	8.19
I14.M1.m0	id	5.64169e-10	8.14
I14.M3.m0	fn	5.57116e-10	8.04
I14.M4.m0	fn	4.98441e-10	7.19
I14.M4.m0	id	3.70538e-10	5.35
I14.M3.m0	id	3.08192e-10	4.45
I14.M6.m0	fn	7.20704e-12	0.10
I14.M6.m0	id	2.82308e-12	0.04
I14.M5.m0	fn	2.52881e-12	0.04
I14.M5.m0	id	1.59239e-12	0.02

Integrated Noise Summary (in V²) Sorted By Noise Contributors
Total Summarized Noise = 6.93211e-09
Total Input Referred Noise = 1.50428e-05
The above noise summary info is for noise data

Parameter	Simulation	Hand calculations
noise density	15.15 nV/ $\sqrt{Hz}$	14.83 nV/ $\sqrt{Hz}$
BW	5 MHz	5 MHz
Thermal rms noise	42.55 μV_{rms}	41.56 μV_{rms}

5T OTA Transient Noise Analysis



■ V_{out} ■ V_{in}

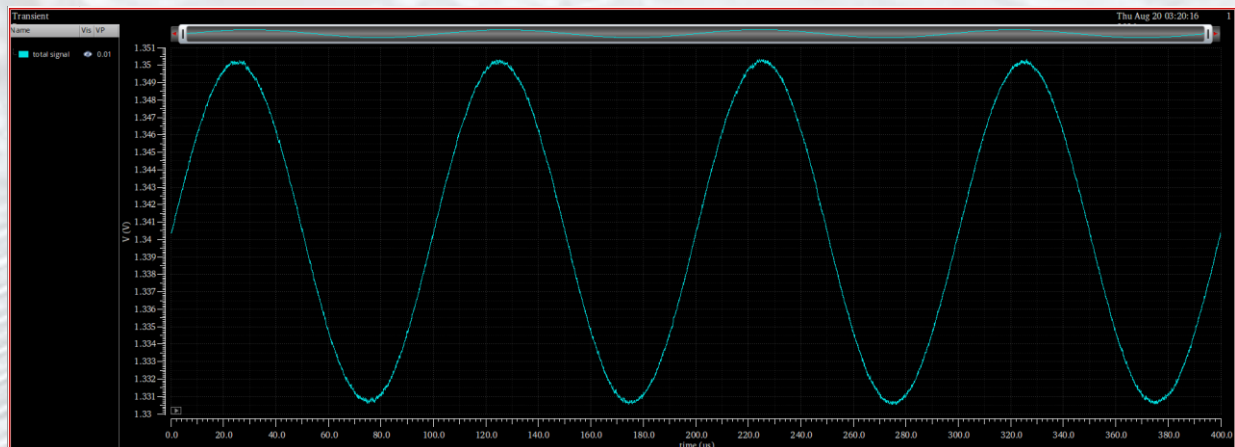
verify that the circuit behaves as a buffer

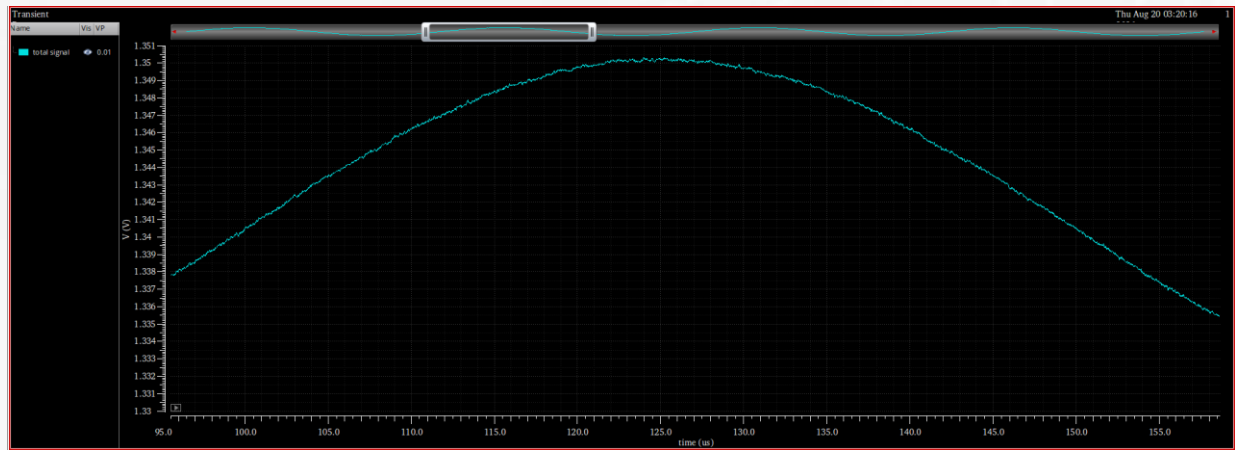
the output follow the input in phase and magnitude with very small DC offset 9.9mv

Now we will run transient noise analysis similar to Part 2. Use a single noise simulation run. Set transient noise upper frequency at 10 times the OTA GBW.

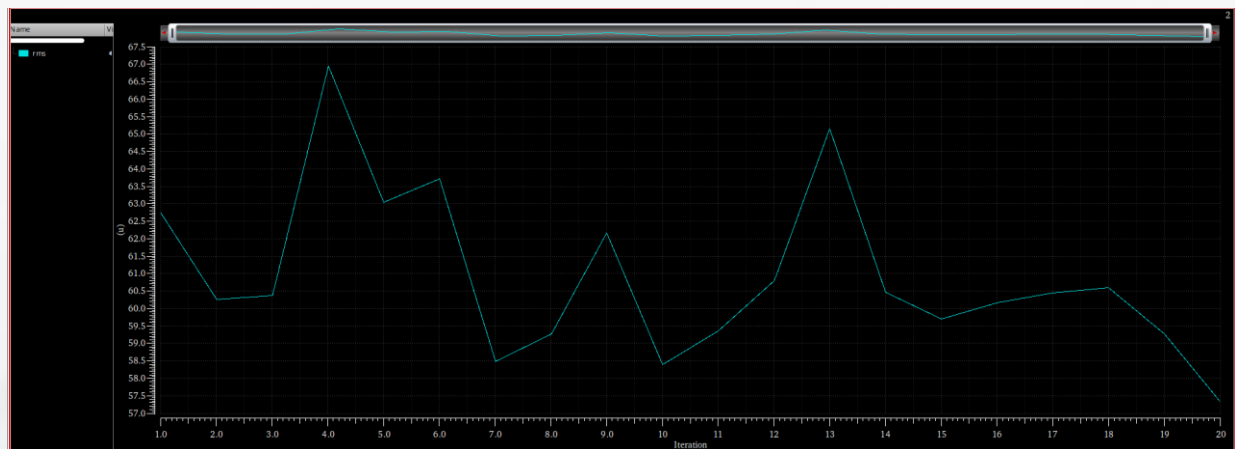
$$f_{max} = \frac{1}{T_{step}} = 10 \times 5MHz = 50MHz \Rightarrow T_{step} = \frac{\tau}{100} = 20ns \Rightarrow \tau = 2\mu s, T_{stop} = 100\tau = 200\mu s$$

$$\Rightarrow f_{min} = \frac{1}{T_{stop}} = 5KHz$$





rms noise vs 20 iteration



Use the calculator to calculate the average rms noise. Compare the calculated value with the rms noise previously obtained in Part 3

The average rms of the total noise (thermal + flicker) in range of (5KHz to 50MHz) for the iterations is $61\mu V_{rms}$

average rms	61u
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In part 3, the thermal noise in the full frequency range is $42.55\mu V_{rms}$

The average rms of the total noise in range of (f_c to 50MHz) (thermal noise dominates) for the iterations is $45\mu V_{rms}$

Expression	Value
1 stddev(value(VT...	45.02E-6

In part 3 from noise summary, the total noise in the full frequency range $\approx \sqrt{6.93 \times 10^{-9}} \approx 83.25\mu V_{rms}$